Spherical Wave Expansion – A brief overview

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1 Introduction

In a source-free region of space, electromagnetic fields can be represented by means of spherical vector wave functions and their associated expansion coefficients. This representation is commonly referred to as the spherical wave expansion and forms a convenient modal description of electromagnetic fields in spherical coordinates.

A widely used formulation of these functions is the power-normalized set of spherical vector wave functions described by J. E. Hansen in *Spherical Near-Field Antenna Measurements*. In this representation, the electromagnetic field outside a spherical surface enclosing the entire source region can be expressed as a superposition of orthogonal spherical modes.

The expansion coefficients can be determined from the electric and/or magnetic field known on such a spherical surface surrounding the source region. Once these coefficients are obtained, the electromagnetic field can be evaluated at arbitrary positions in the surrounding source-free space. This allows both near-field and far-field quantities to be derived from the same modal representation.

The spherical wave expansion is therefore widely used for the analysis of measured or simulated electromagnetic fields, for example when the field distribution of a feed obtained from numerical full-wave simulations is sampled on a spherical surface enclosing the source region. The resulting modal representation can then be used to propagate the electromagnetic field throughout the surrounding source-free space, for example to illuminate subsequent scatterers.

The mathematical formulation and notation used in the following sections are based on the representation given by J. E. Hansen in *Spherical Near-Field Antenna Measurements*.

2 The Spherical Wave Expansion

The electromagnetic field in a source-free region can be expressed as a superposition of spherical vector wave functions. The spherical wave expansion of the electric field can be written as

$$\mathbf{E}(r,\theta,\phi) = \frac{k}{\sqrt{\eta}} \sum_{csmn} Q_{smn}^{(c)} \, \mathbf{F}_{smn}^{(c)}(r,\theta,\phi) \tag{1}$$

$$\mathbf{H}(r,\theta,\phi) = -ik\sqrt{\eta} \sum_{csmn} Q_{smn}^{(c)} \, \mathbf{F}_{3-s,m,n}^{(c)}(r,\theta,\phi) \tag{2}$$

where $\mathbf{F}_{smn}(r,\theta,\phi)$ denote the spherical vector wave functions and Q_{smn} are the corresponding expansion coefficients. The total power of the field can be expressed as

$$P = \frac{1}{2} \sum_{smn} \left| Q_{smn}^{(3)} \right|^2 \text{ watts} \tag{3}$$

The wave number k and the specific admittance η are defined as

$$k = \omega\sqrt{\mu\varepsilon} \tag{4}$$

$$\eta = \sqrt{\frac{\varepsilon}{\mu}} \tag{5}$$

Symbol definitions

$\mathbf{E}(r,\theta,\phi)$	electric field	$\mathbf{H}(r,\theta,\phi)$	magnetic field
(r,θ,ϕ)	spherical coordinates	k	wave number
η	specific admittance	ω	angular frequency
μ	magnetic permeability	ε	electric permittivity
$Q_{smn}^{(c)}$	expansion coefficients	$\mathbf{F}_{smn}^{(c)}$	spherical wave functions

Index ranges and interpretation

Index	Meaning
$c = 1$	spherical Bessel function (finite at origin)
$c = 2$	spherical Neumann function (infinite at origin)
$c = 3$	spherical Hankel function of the first kind (outgoing wave)
$c = 4$	spherical Hankel function of the second kind (incoming wave)
$s = 1, 2$	distinguishes spherical wave functions
$n = 1, 2, 3, \ldots$	polar degree
$m = -n, -n+1, \ldots, 0, \ldots, n-1, n$	azimuthal order

Mode assignment

$Q_{1mn}^{(c)}$: transverse electric (TE)

$Q_{2mn}^{(c)}$: transverse magnetic (TM)

3 The Spherical Wave Expansion Functions

$$\mathbf{F}_{1mn}^{(c)}(r,\theta,\phi) = \frac{1}{\sqrt{2\pi}}\frac{1}{\sqrt{n(n+1)}}\left(-\frac{m}{|m|}\right)^m \left\{ z_n^{(c)}(kr)\frac{im\,\bar{P}_n^{|m|}(\cos\theta)}{\sin\theta}e^{im\phi}\,\hat{\theta} \right.$$
$$\left. -z_n^{(c)}(kr)\frac{d\bar{P}_n^{|m|}(\cos\theta)}{d\theta}e^{im\phi}\,\hat{\phi} \right\} \tag{6}$$

$$\mathbf{F}_{2mn}^{(c)}(r,\theta,\phi) = \frac{1}{\sqrt{2\pi}}\frac{1}{\sqrt{n(n+1)}}\left(-\frac{m}{|m|}\right)^m \left\{ \frac{n(n+1)}{kr}z_n^{(c)}(kr)\bar{P}_n^{|m|}(\cos\theta)e^{im\phi}\,\hat{r} \right.$$
$$+\frac{1}{kr}\frac{d}{d(kr)}\Big(krz_n^{(c)}(kr)\Big)\frac{d\bar{P}_n^{|m|}}{d\theta}e^{im\phi}\,\hat{\theta}$$
$$\left. +\frac{1}{kr}\frac{d}{d(kr)}\Big(krz_n^{(c)}(kr)\Big)\frac{im\bar{P}_n^{|m|}}{\sin\theta}e^{im\phi}\,\hat{\phi} \right\} \tag{7}$$

3.1 Angular Functions

$$\bar{P}_n^{|m|}(\cos\theta) \tag{8}$$

$$\frac{m}{\sin\theta}\bar{P}_n^{|m|}(\cos\theta) \tag{9}$$

$$\frac{d}{d\theta}\bar{P}_n^{|m|}(\cos\theta) \tag{10}$$

Normalized associated Legendre function

$$\bar{P}_n^m(\cos\theta) = \sqrt{\frac{2n+1}{2}\frac{(n-m)!}{(n+m)!}}P_n^m(\cos\theta) \tag{11}$$

Associated Legendre function

$$P_n^m(\cos\theta) = (\sin\theta)^m\frac{d^m P_n(\cos\theta)}{d(\cos\theta)^m} \tag{12}$$

Legendre polynomial

$$P_n(\cos\theta) = \frac{1}{2^n n!}\frac{d^n}{d(\cos\theta)^n}(\cos^2\theta - 1)^n \tag{13}$$

3.2 Radial Functions

$$R_{sn}^{(c)}(kr) = z_n^{(c)}(kr) \tag{14}$$

$$R_{sn}^{(c)}(kr) = \frac{1}{kr}\frac{d}{d(kr)}(krz_n^{(c)}(kr)) \tag{15}$$

Spherical Bessel function

$$z_n^{(1)}(kr) = j_n(kr) \tag{16}$$

Spherical Neumann function

$$z_n^{(2)}(kr) = n_n(kr) \tag{17}$$

Spherical Hankel function of the first kind

$$z_n^{(3)}(kr) = j_n(kr) + i\, n_n(kr) \tag{18}$$

Spherical Hankel function of the second kind

$$z_n^{(4)}(kr) = j_n(kr) - i\, n_n(kr) \tag{19}$$

4 Calculation of the Spherical Wave Expansion Coefficients

The coefficients are obtained by projecting the measured or simulated field onto the corresponding spherical vector wave functions. First, the surface integral over the spherical sampling surface is evaluated. The resulting expression yields the expansion coefficient $Q_{smn}^{(3)}$ for each mode (s, m, n). In the current implementation, the numerical integration and the evaluation of the coefficient expressions are implemented as separate functions. At present, only the outgoing wave case ($c = 3$) is implemented.

$$Q_{smn}^{(3)}\, \tilde{R}_{sn}^{(3,3)}(kA) = \frac{\sqrt{\eta}}{k}(-1)^m \int_0^{2\pi} \int_0^{\pi} \bar{\mathbf{E}}(A, \theta, \phi) \cdot \bar{\mathbf{F}}_{s,-m,n}^{(3)}(A, \theta, \phi) \sin\theta \, d\theta \, d\phi \tag{20}$$

$$\tilde{R}_{sn}^{(c,\gamma)}(kr) = R_{sn}^{(c)}(kr) R_{sn}^{(\gamma)}(kr) + \delta_{s2} n(n+1) \frac{z_n^{(c)}(kr)}{kr} \frac{z_n^{(\gamma)}(kr)}{kr} \tag{21}$$

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \tag{22}$$

δ_{ij} denotes the Kronecker delta. The parameter A denotes the radius of the spherical surface on which the expansion coefficients are evaluated.

5 Implementation notes

Two quadrature schemes are available for the numerical evaluation of the integrals required to determine the spherical wave expansion coefficients: Clenshaw–Curtis quadrature and Gauss–Legendre quadrature.

The number of discretization points is determined by the selected quadrature scheme and the expansion order N and is currently fixed.

For high expansion orders (e.g. $N > 500$), additional speed-up may be achieved by using parallel computing on multi-core CPUs. In such cases, speed-up factors of up to three can be observed.

5.1 Clenshaw–Curtis Quadrature

The Clenshaw–Curtis quadrature includes the poles of the spherical coordinate system, i.e. the sampling grid contains the points $\theta = 0$ and $\theta = \pi$. The discretization in θ is equidistant.

For an expansion order N, the sampling grid consists of $2N + 1$ points in the ϕ direction and $2N + 2$ points in the θ direction.

The advantage of the Clenshaw–Curtis quadrature is the explicit inclusion of the polar points. The disadvantage is the larger number of discretization points required for the numerical integration.

5.2 Gauss–Legendre Quadrature

The Gauss–Legendre quadrature does not include the poles of the spherical coordinate system. The sampling points in the θ direction are not equidistant and correspond to the roots of the Legendre polynomials.

For an expansion order N, the sampling grid consists of $2N + 1$ points in the ϕ direction and $N + 1$ points in the θ direction.

The advantage of the Gauss–Legendre quadrature is the reduced number of discretization points required for the numerical integration. The disadvantages are the absence of sampling points at the poles and the non-equidistant distribution of the θ samples.

5.3 Advices

Due to numerical effects or locally inward-directed power flow in the near-field region, reconstructed field values may become negative. In such cases a logarithmic representation in dB is not possible and the field is displayed on a linear scale in the FieldViewer. The directivity function implemented in the FieldViewer can be applied to both near-field and far-field results.

The current implementation does not perform explicit numerical consistency checks. In particular, no automatic verification of numerical stability or convergence is carried out and no warning messages are generated if numerical instabilities occur. The present implementation should therefore be regarded as an early-stage version and results should be verified by the user when necessary.

In general it is possible to import a field bei generating a grid with the gen_grid() function. Then calculate the spherical components of the E- and H-field on this grid and bring it in the struct form as in the dipole_near() function. After that it can be used for analysis.

For importing Q_{smn} values only a multidimensional array (SxMxN) of the size 2 x (2N+1) x N is needed. This means the full set of coefficients for a given N with the following indexing will be used: S=1:2 , M-> -N=(index 1) : 1 : N=(index end), N=1:N